BUILT — Speed & efficiency pass (v0.56.0)

Five internal optimizations — no new UI, no new surface area. Faster paging, faster lookups, fewer bytes on the wire. All RPS-safe and additive.

TIER 1 — the big wins

Open-PDF LRU cache

- render_page_png re-opened + re-parsed the PDF on EVERY page. Now an LRU of 8 open fitz.Documents is reused.
- PyMuPDF isn't thread-safe -> each doc carries a render lock; the highlight path still uses a fresh doc (it mutates).
- Big win for paging + the loupe.

Verified: LRU reuse + evict + close.

Thread-local DB connections

- db() opened a new SQLite connection + re-applied PRAGMAs on every request (the dossier fires 6+).
- Now a per-thread connection is reused; callers' .close() is a harmless no-op; rebuilds if it goes bad.
- Relaxed/OCR mode still gets a fresh conn (no shared locks).

Verified: reuse, Row-factory, 5-thread x20, 0 errors.

Indexed look-alike (NOCASE)

- Look-Alike scanned the whole parts table via UPPER(name)=UPPER(?).
- Now name/nomenclature are matched with COLLATE NOCASE so the new NOCASE indexes apply.
- EXPLAIN: MULTI-INDEX OR across ix_parts_name + ix_parts_nomenclature on real (selective) data.

Index built by optimize_index.py.

TIER 2 — cheap & worth it

ETag / 304 Not-Modified

- _send now sends an ETag (md5 of the body, stable across gzip). A matching If-None-Match returns 304 with an empty body.
- Repeat views of a page image, the JS, or JSON skip re-sending the bytes entirely — big on slow links.

Verified by curl: 304 on match, 200 on stale, Cache-Control preserved.

Index maintenance + ANALYZE

- optimize_index.py (+ .bat): idempotent CREATE INDEX IF NOT EXISTS for pages(document_id) and parts(name/nomenclature NOCASE), then ANALYZE so the planner uses them.
- EXPLAIN confirms find-in-manual now SEARCHes via ix_pages_document instead of scanning the biggest table.

Run once when OCR is paused (brief write lock; 120s busy-timeout).

SAFE, MEASURED, NO BLOAT

Every change is server-internal — no new pages, no API surface, nothing for the user to learn. All RPS-safe (pure Python, ES-agnostic). 44/44 tests still green; the connection-reuse db() is exercised by the passing pillar suite. The two index builds are the only DB writes and are a one-time, idempotent maintenance step — kept out of the live server so they never contend with the running OCR. This is efficiency by tightening hot paths, not by adding machinery.